NCEA Level 2 Mathematics

91262 · Apply calculus methods in solving problems (Calculus)

Excellence-level practice questions

SOLUTIONS

About these solutions

- These are **full worked solutions** to the 91262 practice excellence questions. Where a question can be done more than one way, an **alternate solution** is also shown.
- They are **not** an official NZQA assessment schedule, and no marks or grades are given. However, solving these levels of questions in the exam should award you an excellence mark.

How to use them

- Attempt each question properly first, then check your final answer against the answer schedule.
- Only read a solution once you're stuck or have finished, and don't just copy it. Close it and try the question again on your own.
- Your method doesn't have to match the one shown, as long as it is correct and fully explained.

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- If you spot an error or typo, please let me know so I can fix it.

Good luck!

Assessment Schedule – Practice Excellence Questions

Level 2 Mathematics: Calculus (91262)

Evidence Statement

Q	Solution
1	Surface area (base and four sides): $x^2 + 4xh = 300 \Rightarrow h = \dfrac{300 - x^2}{4x}$ $V = x^2 h = x^2\left(\dfrac{300 - x^2}{4x}\right) = 75x - \dfrac{x^3}{4}$ $\dfrac{dV}{dx} = 75 - \dfrac{3x^2}{4} = 0$ $x^2 = 100 \Rightarrow x = 10 \quad (x > 0)$ $h = \dfrac{300 - 100}{40} = 5$, so $V = 10^2 \times 5 = 500\ \text{cm}^3$ Maximum: at $x = 9$, $\dfrac{dV}{dx} = 14.25 > 0$; at $x = 11$, $\dfrac{dV}{dx} = -15.75 < 0$. The gradient changes from positive to negative, so it is a maximum. **Alternate solution:** Justify with the second derivative: $\dfrac{d^2V}{dx^2} = -\dfrac{3x}{2} = -15 < 0$ at $x = 10$, so it is a maximum.
2	Let the height of the rectangle be h. The width is $2r$. Perimeter: $2r + 2h + \pi r = 6 \Rightarrow h = \dfrac{6 - 2r - \pi r}{2}$ $A = 2rh + \frac{1}{2}\pi r^2$ $= r(6 - 2r - \pi r) + \frac{1}{2}\pi r^2$ $= 6r - 2r^2 - \frac{1}{2}\pi r^2$ $\dfrac{dA}{dr} = 6 - 4r - \pi r = 0$ $r = \dfrac{6}{4 + \pi} \approx 0.840\ \text{m}$ $A = 6r - \left(2 + \frac{\pi}{2}\right) r^2 = \dfrac{36}{4 + \pi} - \dfrac{18}{4 + \pi} = \dfrac{18}{4 + \pi} \approx 2.52\ \text{m}^2$ $\dfrac{d^2A}{dr^2} = -(4 + \pi) < 0$, so this is a maximum. **Alternate solution:** $A = 6r - \left(2 + \frac{\pi}{2}\right) r^2$ is a quadratic in r with a negative r^2 coefficient, so its maximum is at the vertex: $r = \dfrac{-6}{2 \times -\left(2 + \frac{\pi}{2}\right)} = \dfrac{6}{4 + \pi}$
3	Let the right-hand vertices be $(x, 0)$ and $(x, 12 - x^2)$, with $0 < x < \sqrt{12}$. $A = 2x(12 - x^2) = 24x - 2x^3$ $\dfrac{dA}{dx} = 24 - 6x^2 = 0 \Rightarrow x = 2$ $A = 4 \times 8 = 32\ \text{units}^2$ $\dfrac{d^2A}{dx^2} = -12x = -24 < 0$, so this is a maximum. **Alternate solution:** Justify with the gradient: at $x = 1$, $\dfrac{dA}{dx} = 18 > 0$; at $x = 3$, $\dfrac{dA}{dx} = -30 < 0$. Positive to negative, so a maximum.

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Q	Solution
4	Put the right angle at the origin, with the 8 cm side along the x-axis and the 6 cm side along the y-axis. The hypotenuse joins $(8, 0)$ and $(0, 6)$: $y = 6 - \frac{3}{4}x$ The rectangle is x by y, where (x, y) is on the hypotenuse: $A = x\left(6 - \frac{3}{4}x\right) = 6x - \frac{3}{4}x^2$ $\frac{dA}{dx} = 6 - \frac{3}{2}x = 0 \Rightarrow x = 4,\ y = 3$ $A = 12$ cm^2 $\frac{d^2A}{dx^2} = -\frac{3}{2} < 0$, so this is a maximum. **Alternate solution:** Use similar triangles: the small triangle above the rectangle is similar to the whole triangle, so $\frac{y}{6} = \frac{8 - x}{8}$, giving $y = 6 - \frac{3}{4}x$ as above.
5	Let the width be w m (parallel to the internal fences) and the length be L m. There are 4 widths (2 sides and 2 internal fences) and 2 lengths: $4w + 2L = 240 \Rightarrow L = 120 - 2w$ $A = wL = 120w - 2w^2$ $\frac{dA}{dw} = 120 - 4w = 0 \Rightarrow w = 30,\ L = 60$ Width 30 m, length 60 m, maximum area = 1800 m^2 $\frac{d^2A}{dw^2} = -4 < 0$, so this is a maximum.
6	$x = 10 - 2y$. Let $S = x^2 + y^2$: $S = (10 - 2y)^2 + y^2 = 100 - 40y + 5y^2$ $\frac{dS}{dy} = -40 + 10y = 0 \Rightarrow y = 4,\ x = 2$ $S = 4 + 16 = 20$ $\frac{d^2S}{dy^2} = 10 > 0$, so the minimum value is 20. **Alternate solution:** Write S in terms of x instead: $y = \frac{10 - x}{2}$ $S = x^2 + \frac{(10 - x)^2}{4}$ $\frac{dS}{dx} = 2x - \frac{10 - x}{2} = 0 \Rightarrow x = 2,\ y = 4,\ S = 20$

Q	Solution
7	Let the side material cost \$$c$ per cm^2, so the top and bottom cost \$$2c$ per cm^2. Volume: $\pi r^2 h = 1000 \Rightarrow h = \dfrac{1000}{\pi r^2}$ $C = 2c\left(2\pi r^2\right) + c\,(2\pi r h)$ $= c\left(4\pi r^2 + 2\pi r \times \dfrac{1000}{\pi r^2}\right) = c\left(4\pi r^2 + \dfrac{2000}{r}\right)$ $\dfrac{dC}{dr} = c\left(8\pi r - \dfrac{2000}{r^2}\right) = 0$ $r^3 = \dfrac{250}{\pi} \Rightarrow r = \sqrt[3]{\dfrac{250}{\pi}} \approx 4.30$ cm $\dfrac{d^2C}{dr^2} = c\left(8\pi + \dfrac{4000}{r^3}\right) > 0$, so this is a minimum.
8	$v = \int (6t - 12)\,dt = 3t^2 - 12t + c$ $t = 0,\ v = 9 \Rightarrow c = 9$ $s = \int (3t^2 - 12t + 9)\,dt = t^3 - 6t^2 + 9t + d$ $t = 0,\ s = 0 \Rightarrow d = 0$ $s(4) = 64 - 96 + 36 = 4$ The particle is 4 m from O (in the positive direction).
9	Let t be the time (s) after the truck passes the lights. Truck: $s_T = 12t$ Motorbike (for $t \geq 5$): $a = 2 \Rightarrow v = 2t + c$, and $v = 0$ at $t = 5$, so $v = 2(t - 5)$. $s = (t - 5)^2 + d$, and $s = 0$ at $t = 5$, so $s_M = (t - 5)^2$. $(t - 5)^2 = 12t$ $t^2 - 22t + 25 = 0$ $t = 11 \pm \sqrt{96} \Rightarrow t = 11 + 4\sqrt{6} \approx 20.80 \quad (t > 5)$ Distance: $s = 12t \approx 249.6$ m (about 250 m from the lights) Speed: $v = 2(t - 5) \approx 31.6\ \text{m s}^{-1}$ **Alternate solution:** Let T be the time after the motorbike starts. Then $s_M = T^2$, $v_M = 2T$, and the truck has a 60 m head start: $s_T = 12T + 60$. $T^2 = 12T + 60$ $T^2 - 12T - 60 = 0 \Rightarrow T = 6 + \sqrt{96} \approx 15.80 \quad (T > 0)$ $s = T^2 \approx 249.6$ m, $\quad v = 2T \approx 31.6\ \text{m s}^{-1}$

Q	Solution
10	Take upwards as positive, with the origin at the roof. $v = 20 - 9.8t$ $s = 20t - 4.9t^2$ The ground is 25 m below the roof, so $s = -25$: $20t - 4.9t^2 = -25$ $4.9t^2 - 20t - 25 = 0$ $t = \dfrac{20 + \sqrt{890}}{9.8} \approx 5.085$ s $\quad (t > 0)$ $v = 20 - 9.8(5.085) \approx -29.8$ The ball hits the ground at a speed of about 29.8 m s^{-1}. **Alternate solution:** Find the maximum height first. $v = 0$ when $t = \dfrac{20}{9.8} \approx 2.041$ s, and then $s = 20(2.041) - 4.9(2.041)^2 \approx 20.41$ m above the roof. The ball then falls from rest through $20.41 + 25 = 45.41$ m, with $s = 4.9T^2$ and $v = 9.8T$: $4.9T^2 = 45.41 \Rightarrow T \approx 3.044$ s, $\quad v = 9.8 \times 3.044 \approx 29.8$ m s^{-1}
11 (i)	Reaction distance: $25 \times 0.8 = 20$ m Braking (from when the brakes are applied): $v = 25 - 5t$, $\quad v = 0$ when $t = 5$ $s = 25t - 2.5t^2 \Rightarrow s(5) = 125 - 62.5 = 62.5$ m Total stopping distance = 20 + 62.5 = 82.5 m > 80 m, so the car does **not** stop before the hazard.
(ii)	The brakes have 80 − 20 = 60 m to stop the car. Let the deceleration be a: $v = 25 - at$, $\quad v = 0$ when $t = \dfrac{25}{a}$ $s = 25t - \frac{1}{2}at^2$ $s\left(\frac{25}{a}\right) = \dfrac{625}{a} - \dfrac{625}{2a} = \dfrac{625}{2a} = 60$ $a = \dfrac{625}{120} = \dfrac{125}{24} \approx 5.21$ m s^{-2}
12	The tangent has gradient 4: $\dfrac{dy}{dx} = 3x^2 - 4x = 4$ $3x^2 - 4x - 4 = 0$ $(3x + 2)(x - 2) = 0 \Rightarrow x = 2$ or $x = -\frac{2}{3}$ $x = 2$: $y = 8 - 8 + 1 = 1$, so $1 = 8 + c \Rightarrow c = -7$ $x = -\frac{2}{3}$: $y = -\frac{8}{27} - \frac{8}{9} + 1 = -\frac{5}{27}$, so $-\frac{5}{27} = -\frac{8}{3} + c \Rightarrow c = \frac{67}{27}$

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Q	Solution
13	Let the tangent touch the curve at $x = a$. Gradient $2a - 4$; point $(a,\ a^2 - 4a + 5)$. The tangent passes through $(1, -2)$: $-2 - (a^2 - 4a + 5) = (2a - 4)(1 - a)$ $-a^2 + 4a - 7 = -2a^2 + 6a - 4$ $a^2 - 2a - 3 = 0 \Rightarrow (a - 3)(a + 1) = 0$ $a = 3$: gradient 2, point $(3, 2)$: $y - 2 = 2(x - 3) \Rightarrow y = 2x - 4$ $a = -1$: gradient -6, point $(-1, 10)$: $y - 10 = -6(x + 1) \Rightarrow y = -6x + 4$ **Alternate solution:** A line through $(1, -2)$ is $y = m(x - 1) - 2$. Substitute into the curve: $x^2 - 4x + 5 = mx - m - 2$ $x^2 - (4 + m)x + (7 + m) = 0$ Tangent, so $\Delta = 0$: $(4 + m)^2 - 4(7 + m) = 0$ $m^2 + 4m - 12 = 0 \Rightarrow m = 2$ or $m = -6$ $y = 2x - 4$ and $y = -6x + 4$
14	At $x = -1$: $y = -1 - 3 + 2 = -2$, $\dfrac{dy}{dx} = 3x^2 - 6x = 9$ Tangent: $y + 2 = 9(x + 1) \Rightarrow y = 9x + 7$ Where the tangent meets the curve: $x^3 - 3x^2 + 2 = 9x + 7$ $x^3 - 3x^2 - 9x - 5 = 0$ The line touches the curve at $x = -1$, so $x = -1$ is a repeated root and $(x + 1)^2$ is a factor. Matching the constant term, $(x + 1)^2(x - p)$ has constant $-p = -5$, so $x^3 - 3x^2 - 9x - 5 = (x + 1)^2(x - 5)$ $x = 5$, $y = 9(5) + 7 = 52$. P$(5, 52)$
15	$f(0) = 0 \Rightarrow d = 0$ $f'(x) = 3ax^2 + 2bx + c$, $f'(0) = 3 \Rightarrow c = 3$ Stationary point at $(2, -4)$: $f'(2) = 0$: $12a + 4b + 3 = 0$ $f(2) = -4$: $8a + 4b + 6 = -4 \Rightarrow 8a + 4b = -10$ Subtract: $4a = 7 \Rightarrow a = \frac{7}{4}$, $b = -6$ $a = \frac{7}{4}$, $b = -6$, $c = 3$, $d = 0$ Nature: $f'(x) = \frac{21}{4}x^2 - 12x + 3$. $f'(1) = -3.75 < 0$ and $f'(3) = 14.25 > 0$. Negative to positive, so $(2, -4)$ is a **local minimum**. **Alternate solution:** Nature with the second derivative: $f''(x) = \frac{21}{2}x - 12$, $f''(2) = 9 > 0$, so it is a local minimum.

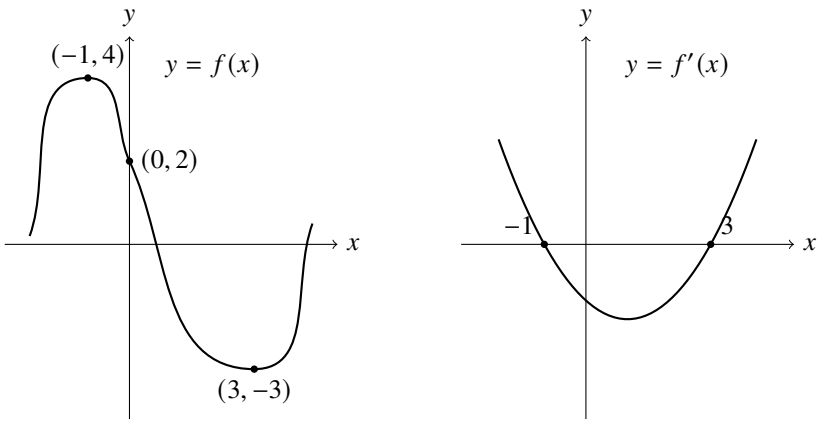
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Q	Solution
16	$\frac{dy}{dx} = 3x^2 + 2px + q$ $\frac{dy}{dx} = 0$ at $x = 1$: $\quad 3 + 2p + q = 0$ $(1, 2)$ is on the curve: $\quad 1 + p + q + 4 = 2 \Rightarrow p + q = -3$ Subtract: $\quad 3 + p = 3 \Rightarrow p = 0,\ q = -3$ $y = x^3 - 3x + 4,\ \frac{dy}{dx} = 3x^2 - 3 = 0 \Rightarrow x = \pm 1$ Other stationary point: $x = -1,\ y = -1 + 3 + 4 = 6,$ i.e. $(-1, 6)$ $\frac{dy}{dx}$ at $x = -2$ is $9 > 0$, and at $x = 0$ is $-3 < 0$: positive to negative, so a **local maximum**. **Alternate solution:** Nature with the second derivative: $\frac{d^2y}{dx^2} = 6x = -6 < 0$ at $x = -1$, so it is a local maximum.
17	Turning point at $x = 2$: $3(4) + 2k - 4 = 0 \Rightarrow k = -4$ $y = \int (3x^2 - 4x - 4)\, dx = x^3 - 2x^2 - 4x + c$ $(0, 5) \Rightarrow c = 5$ $y = x^3 - 2x^2 - 4x + 5$ Other turning point: $3x^2 - 4x - 4 = (3x + 2)(x - 2) = 0 \Rightarrow x = -\frac{2}{3}$ $y = -\frac{8}{27} - \frac{8}{9} + \frac{8}{3} + 5 = \frac{175}{27} \quad \Rightarrow \left(-\frac{2}{3},\ \frac{175}{27}\right)$
18	$f'(x) = 3x^2 + 2kx + 12$. For f to be increasing for all x, we need $f'(x) \geq 0$ for all x. $f'(x)$ is a parabola opening upwards, so it must not cross the x-axis: $\Delta \leq 0$. $(2k)^2 - 4(3)(12) \leq 0$ $4k^2 \leq 144$ $-6 \leq k \leq 6$ *Note:* at $k = \pm 6$, $f'(x) = 3(x \pm 2)^2$ is zero at only one point (a stationary point of inflection), and the function is still increasing. If the question is read as "no stationary points at all", the answer is $-6 < k < 6$. **Alternate solution:** Complete the square: $f'(x) = 3\left(x + \frac{k}{3}\right)^2 + 12 - \frac{k^2}{3}$ The minimum value of $f'(x)$ is $12 - \frac{k^2}{3}$. This must be ≥ 0, so $k^2 \leq 36$: $-6 \leq k \leq 6$.
19	$h = 2r$, so $V = \frac{1}{3}\pi r^2(2r) = \frac{2}{3}\pi r^3$ $\frac{dV}{dr} = 2\pi r^2$ When $V = 18\pi$: $\frac{2}{3}\pi r^3 = 18\pi \Rightarrow r^3 = 27 \Rightarrow r = 3$ $\frac{dV}{dr} = 2\pi(3)^2 = 18\pi$ cm^3 per cm
20 (i)	Decreasing where $\frac{dy}{dx} < 0$: $3x^2 - 3k^2 < 0$ $x^2 < k^2$ $-k < x < k \qquad (k > 0)$

Q	Solution
(ii)	$\frac{dy}{dx} = 0$ at $x = \pm k$. The gradient changes from negative to positive at $x = k$, so the local minimum is at $x = k$. On the x-axis: $k^3 - 3k^2(k) + 4 = 0$ $-2k^3 + 4 = 0$ $k^3 = 2 \Rightarrow k = \sqrt[3]{2}$
21	$f(x+h) = (x+h)^3 - 2(x+h) = x^3 + 3x^2h + 3xh^2 + h^3 - 2x - 2h$ $f(x+h) - f(x) = 3x^2h + 3xh^2 + h^3 - 2h$ $\frac{f(x+h) - f(x)}{h} = 3x^2 + 3xh + h^2 - 2$ $f'(x) = \lim_{h\to 0}\left(3x^2 + 3xh + h^2 - 2\right) = 3x^2 - 2$
22	$\frac{dy}{dx} = 2x - 4$, so the tangent gradient at $x = 1$ is -2 and the normal gradient is $\frac{1}{2}$. $y + 3 = \frac{1}{2}(x - 1) \Rightarrow y = \frac{1}{2}x - \frac{7}{2}$ Where the normal meets the curve: $x^2 - 4x = \frac{1}{2}x - \frac{7}{2}$ $2x^2 - 9x + 7 = 0$ $(2x - 7)(x - 1) = 0$ $x = \frac{7}{2}$, $y = \frac{49}{4} - 14 = -\frac{7}{4}$. Other point: $\left(\frac{7}{2}, -\frac{7}{4}\right)$
23	At $x = 2$: $y = 4$, $\frac{dy}{dx} = -2x = -4$ $y - 4 = -4(x - 2) \Rightarrow y = 12 - 4x$ Intercepts: $(0, 12)$ and $(3, 0)$ Area $= \frac{1}{2} \times 3 \times 12 = 18$ units^2
24	The gradient is $g(x) = \frac{dy}{dx} = 3x^2 - 12x + 9$. Find where g is smallest: $\frac{dg}{dx} = 6x - 12 = 0 \Rightarrow x = 2$ $g(2) = 12 - 24 + 9 = -3$ g is a parabola opening upwards, so this is its minimum. $y = 8 - 24 + 18 + 1 = 3$. Smallest gradient -3, at the point $(2, 3)$. **Alternate solution:** Complete the square: $g(x) = 3(x^2 - 4x) + 9 = 3(x-2)^2 - 3$. Since $(x-2)^2 \geq 0$, the smallest gradient is -3, when $x = 2$.
25 (i)	$v_A = 2t - 4$, $v_B = 6 - 2t$ $2t - 4 = 6 - 2t \Rightarrow t = 2.5$ s $s_A = 6.25 - 10 = -3.75$ m, $s_B = 15 - 6.25 = 8.75$ m Distance apart $= 8.75 - (-3.75) = 12.5$ m
(ii)	$t^2 - 4t = 6t - t^2$ $2t^2 - 10t = 0 \Rightarrow 2t(t - 5) = 0$ $t = 0$ is the start, so they are next at the same position when $t = 5$ s.

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Q	Solution
26	Let P be (x, x^2): $d^2 = (x-16)^2 + \left(x^2 - \frac{1}{2}\right)^2$ $= x^2 - 32x + 256 + x^4 - x^2 + \frac{1}{4}$ $= x^4 - 32x + 256.25$ $\frac{d(d^2)}{dx} = 4x^3 - 32 = 0 \Rightarrow x^3 = 8 \Rightarrow x = 2$ P$(2, 4)$. $\frac{d^2(d^2)}{dx^2} = 12x^2 = 48 > 0$, so this is a minimum. $d^2 = 16 - 64 + 256.25 = 208.25 = \frac{833}{4} \Rightarrow d = \frac{7\sqrt{17}}{2} \approx 14.4$
27	$P = 155x - (x^3 - 24x^2 + 200x) = -x^3 + 24x^2 - 45x$ $\frac{dP}{dx} = -3x^2 + 48x - 45 = -3(x-1)(x-15) = 0$ $x = 1$ or $x = 15$ $\frac{d^2P}{dx^2} = -6x + 48$: at $x = 15$ it is $-42 < 0$ (maximum); at $x = 1$ it is $42 > 0$ (minimum). $P(15) = -3375 + 5400 - 675 = 1350$ Make 15 items per day, for a maximum profit of \$1350. **Alternate solution:** Justify with the gradient: at $x = 14$, $\frac{dP}{dx} = 39 > 0$; at $x = 16$, $\frac{dP}{dx} = -45 < 0$. Positive to negative, so a maximum.
28	Let the cylinder have radius r and height h. In the cross-section, similar triangles give $\frac{12-h}{r} = \frac{12}{6} \Rightarrow h = 12 - 2r$ $V = \pi r^2(12 - 2r) = \pi\left(12r^2 - 2r^3\right)$ $\frac{dV}{dr} = \pi\left(24r - 6r^2\right) = 6\pi r(4 - r) = 0 \Rightarrow r = 4 \quad (r > 0)$ $V = \pi(16)(4) = 64\pi \approx 201$ cm^3 $\frac{d^2V}{dr^2} = \pi(24 - 12r) = -24\pi < 0$, so this is a maximum. **Alternate solution:** Use the equation of the slant edge: put the cone's axis on the y-axis with the base on the x-axis. The edge joins $(6, 0)$ and $(0, 12)$, so $y = 12 - 2x$. The top corner of the cylinder is (r, h) on this line, so $h = 12 - 2r$.
29	$f''(x) = 6x - 4$ $f'(x) = 3x^2 - 4x + c$ Turning point at $x = 1$: $3 - 4 + c = 0 \Rightarrow c = 1$ $f(x) = x^3 - 2x^2 + x + d$ $f(1) = 2$: $1 - 2 + 1 + d = 2 \Rightarrow d = 2$ $y = x^3 - 2x^2 + x + 2$

Q	Solution
30	 $f(x)$: local maximum at $(-1, 4)$, passes through $(0, 2)$, local minimum at $(3, -3)$. $f'(x)$ (one possible shape): • $f'(-1) = f'(3) = 0$: the gradient graph cuts the x-axis at $x = -1$ and $x = 3$, where f has its turning points. • $f'(x) > 0$ for $x < -1$ and $x > 3$: the gradient graph is above the axis there, where f is increasing. • $f'(x) < 0$ for $-1 < x < 3$: the gradient graph is below the axis there, where f is decreasing. The values $f(-1) = 4$, $f(0) = 2$ and $f(3) = -3$ only affect the graph of f, not f'.
31	$\dfrac{dy}{dx} = 3x^2 + 3$ $x^2 \geq 0$ for all x, so $3x^2 + 3 \geq 3$: no tangent has a gradient less than 3. The smallest gradient, 3, occurs when $x = 0$, where $y = 0$. Tangent: $y = 3x$
32	$\dfrac{dy}{dx} = 2x$, so the tangents have gradients $2a$ and $2b$. At A: $y - a^2 = 2a(x - a) \Rightarrow y = 2ax - a^2$ At B: $y = 2bx - b^2$ Perpendicular: $(2a)(2b) = -1 \Rightarrow ab = -\frac{1}{4}$ Where the tangents meet ($a \neq b$): $2ax - a^2 = 2bx - b^2$ $2x(a - b) = (a - b)(a + b) \Rightarrow x = \dfrac{a + b}{2}$ $y = 2a\left(\dfrac{a + b}{2}\right) - a^2 = ab = -\frac{1}{4}$ So the tangents always meet on the line $y = -\frac{1}{4}$.
33	Side length $= \dfrac{P}{4}$, so $A = \left(\dfrac{P}{4}\right)^2 = \dfrac{P^2}{16}$ $\dfrac{dA}{dP} = \dfrac{P}{8} = \dfrac{20}{8} = 2.5$ cm^2 per cm

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Q	Solution
34	The base of the gutter is $30 - 2x$ cm. $A = x(30 - 2x) = 30x - 2x^2$ $\frac{dA}{dx} = 30 - 4x = 0 \Rightarrow x = 7.5$ cm $\frac{d^2A}{dx^2} = -4 < 0$, so a maximum (area $= 7.5 \times 15 = 112.5$ cm^2). **Alternate solution:** $A = 2x(15 - x)$ is a parabola with zeros at $x = 0$ and $x = 15$, opening downwards. Its maximum is halfway between the zeros, at $x = 7.5$ cm.
35	Let $y = x^3 - 3x^2$. $\frac{dy}{dx} = 3x^2 - 6x = 3x(x - 2) = 0 \Rightarrow x = 0$ or $x = 2$ $x = 0$: $y = 0$, and $\frac{d^2y}{dx^2} = 6x - 6 = -6 < 0$, so $(0, 0)$ is a local maximum. $x = 2$: $y = -4$, and $\frac{d^2y}{dx^2} = 6 > 0$, so $(2, -4)$ is a local minimum. The solutions of $x^3 - 3x^2 = k$ are where the horizontal line $y = k$ meets the cubic. The line cuts the curve three times only when it lies strictly between the local minimum and the local maximum: $-4 < k < 0$ (At $k = 0$ or $k = -4$ the line touches a turning point, giving only two distinct solutions.)
36	At $x = 0$: $y = 0$. At $x = 3$: $y = 27 - 3 = 24$. Gradient of chord $= \frac{24 - 0}{3 - 0} = 8$ $\frac{dy}{dx} = 3x^2 - 1 = 8$ $x^2 = 3 \Rightarrow x = \sqrt{3}$ (reject $-\sqrt{3}$, not between 0 and 3) $y = 3\sqrt{3} - \sqrt{3} = 2\sqrt{3}$ The point is $\left(\sqrt{3},\ 2\sqrt{3}\right)$.
37	Stage 1 (accelerating): $v = 1.2t$, $s = 0.6t^2$. At $t = 5$: $v = 6$ m s^{-1}, $s = 15$ m. Stage 3 (decelerating, with t measured from the start of this stage): $v = 6 - 1.5t$, $s = 6t - 0.75t^2$. $v = 0$ when $t = 4$, so $s = 24 - 12 = 12$ m. Stage 2 (constant speed): distance $= 60 - 15 - 12 = 33$ m at 6 m s^{-1}. time $= \frac{33}{6} = 5.5$ s

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Q	Solution
38	Maximum turning point $(3, 2)$, so $f(x) = a(x-3)^2 + 2$ with $a < 0$. $f'(x) = 2a(x-3)$, so $f'(1) = -4a$, and $f(1) = 4a + 2$. Tangent at $x = 1$: $y - (4a+2) = -4a(x-1)$. It passes through $(0, 0)$: $-(4a+2) = 4a$ $8a = -2 \Rightarrow a = -\frac{1}{4}$ $f(x) = -\frac{1}{4}(x-3)^2 + 2 \left(= -\frac{1}{4}x^2 + \frac{3}{2}x - \frac{1}{4}\right)$ **Alternate solution:** Let $f(x) = ax^2 + bx + c$: $f'(3) = 0: \quad 6a + b = 0$ $f(3) = 2: \quad 9a + 3b + c = 2$ Tangent at $x = 1$ through the origin: $f'(1) = \dfrac{f(1) - 0}{1 - 0}$ $2a + b = a + b + c \Rightarrow c = a$ Then $b = -6a$ and $9a - 18a + a = 2 \Rightarrow a = -\frac{1}{4}$, $b = \frac{3}{2}$, $c = -\frac{1}{4}$.
39	$C = 6v^2 + \dfrac{12\,000}{v}$ $\dfrac{dC}{dv} = 12v - \dfrac{12\,000}{v^2} = 0$ $v^3 = 1000 \Rightarrow v = 10 \text{ km h}^{-1}$ $C = 6(100) + 1200 = \$1800$ $\dfrac{d^2C}{dv^2} = 12 + \dfrac{24\,000}{v^3} = 36 > 0$, so this is a minimum. **Alternate solution:** Justify with the gradient: at $v = 9$, $\dfrac{dC}{dv} \approx -40.1 < 0$; at $v = 11$, $\dfrac{dC}{dv} \approx 32.8 > 0$. Negative to positive, so a minimum.
40 (i)	$s = \displaystyle\int (12 - 3t^2)\, dt = 12t - t^3 + c, \quad s = 0 \text{ at } t = 0 \Rightarrow c = 0$ $v = 0: \quad 12 - 3t^2 = 0 \Rightarrow t = 2 \quad (t > 0)$ $s(2) = 24 - 8 = 16 \text{ m}$ $v > 0$ before $t = 2$ and $v < 0$ after, so the particle is furthest from O in the positive direction at $t = 2$: 16 m.
(ii)	$12t - t^3 = 0$ $t(12 - t^2) = 0 \Rightarrow t = 2\sqrt{3} \approx 3.46 \text{ s} \quad (t > 0)$

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